

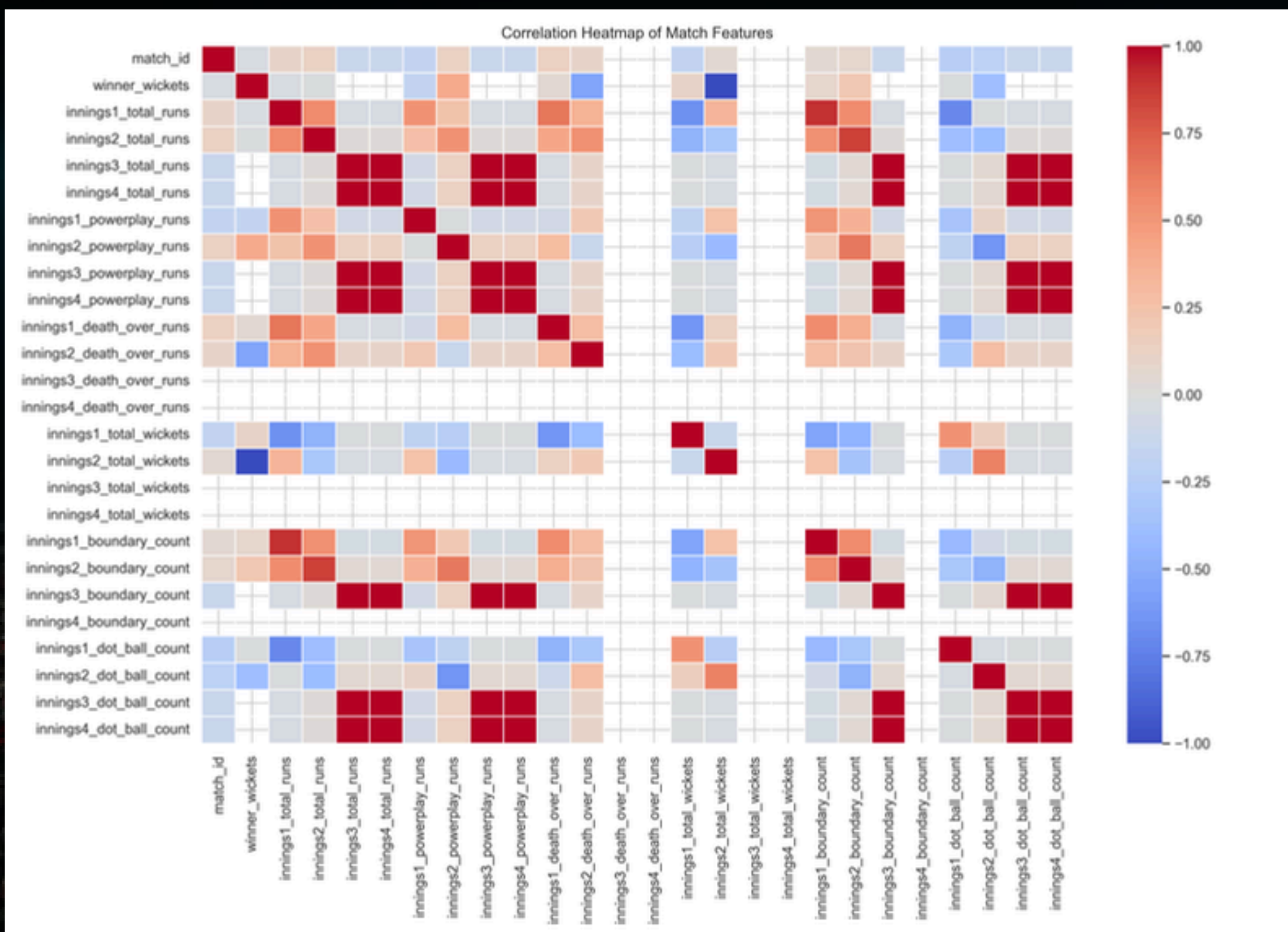
NPL 2025: Match Outcome Prediction

Team Rajesh · June 26, 2026

Problem Statement

Which pre-match and in-game statistical factors—including toss outcome, powerplay performance, bowling economy, and phase-wise run scoring—are significant predictors of match outcome (win/loss) in the Nepal Premier League 2025?

Data Sources & Format



Raw Input

64 Cricsheet-style CSV files in the repo root, each capturing a full match in ball-by-ball format.

- **Info lines** — match metadata: date, teams, toss decision, winner
- **Ball lines** — over-by-over events: batsman, bowler, runs, wickets

Outputs Produced

- matches.csv — one row per match
- deliveries.csv — one row per ball
- match_features.csv — aggregated features for modeling

ETL – data_loader.py

Converts raw Cricsheet CSVs into two tidy, analysis-ready tables with robust handling of missing values.



Parse Files

Build
DataFrames

Save Outputs

The pipeline flows from parsing raw CSV files through building validated DataFrames to saving structured outputs — ensuring data quality at every stage.

parse_match_file

Extracts match metadata: date, teams, toss, and winner.

Delivery Extraction

Per-ball rows with match_id, innings, over_ball, batsman, bowler, runs, and wicket info.

load_data

Coerces numeric columns, validates, and saves CSVs. Raises an error if no CSVs are found.

Exploratory Data Analysis — eda.py

Transformation Steps

- Convert deliveries to numeric types
- Derive per-innings aggregates: total runs, powerplay, death overs, wickets, boundaries, dot balls
- Pivot innings to wide format and merge with match metadata
- Output match_features.csv for modeling

Diagnostic Visualizations

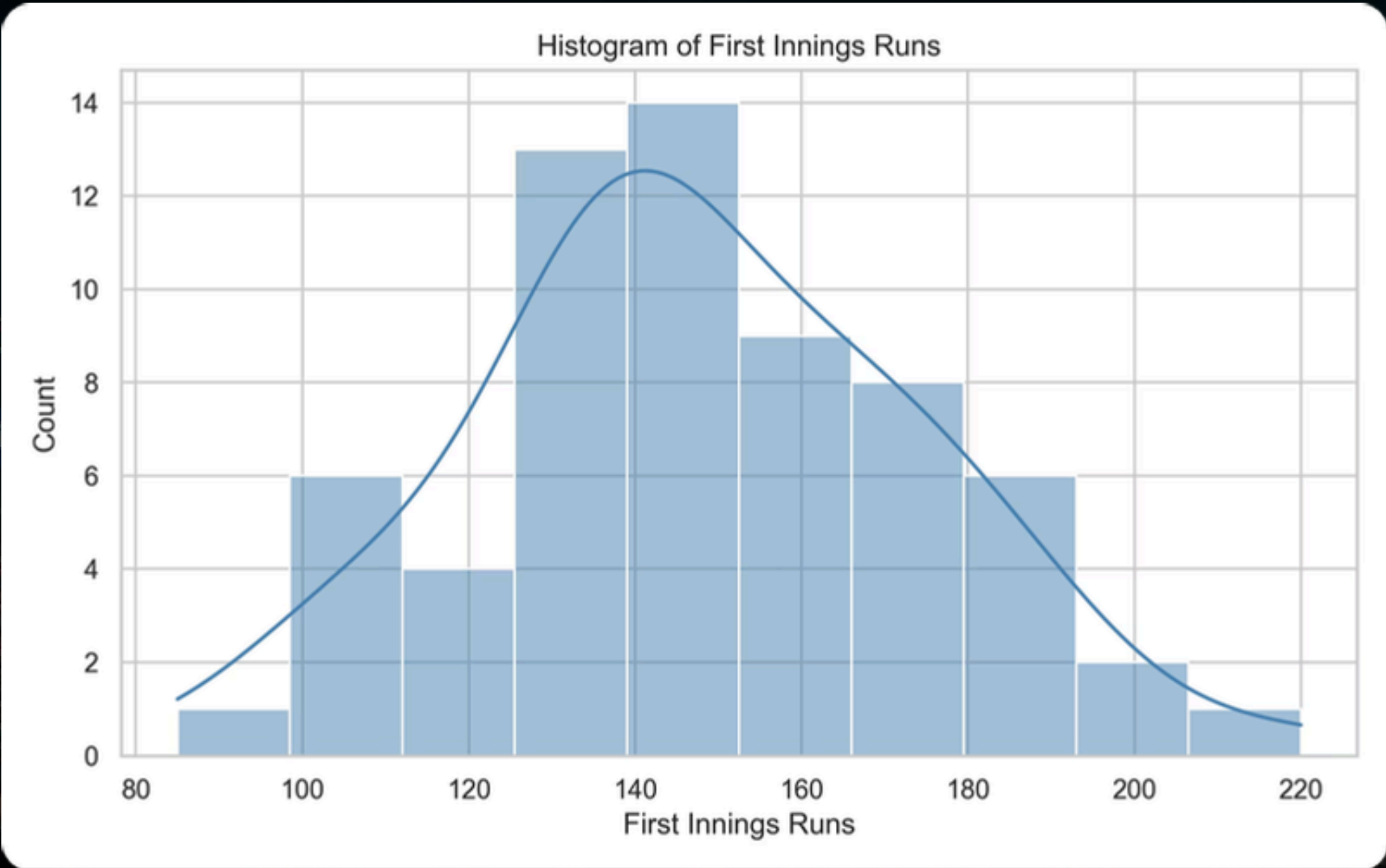
- Histogram, boxplot, scatter, correlation heatmap
- Toss decision bar chart and wicket distribution pie chart



Our Hypothesis

Hypothesis	Test Used	Statistic	p-value	Decision
H1: Does winning the toss give a statistically significant advantage?	Chi-Square	94.69	0.0076	Supported
H2: Do first-innings runs differ across teams' winning probabilities?	One-Way ANOVA	1.7827	0.1008	Rejected
H3: Can we build a model that accurately classifies match winners based on available features?	LR Likelihood Ratio	9.8955	0.1291	Rejected
H4: Which batting and bowling metrics (strike rate, economy rate, powerplay score) are strongest predictors of match outcome?	LR Coefficient Test	-1.8239	0.0682	Rejected

First Innings Runs - Distribution

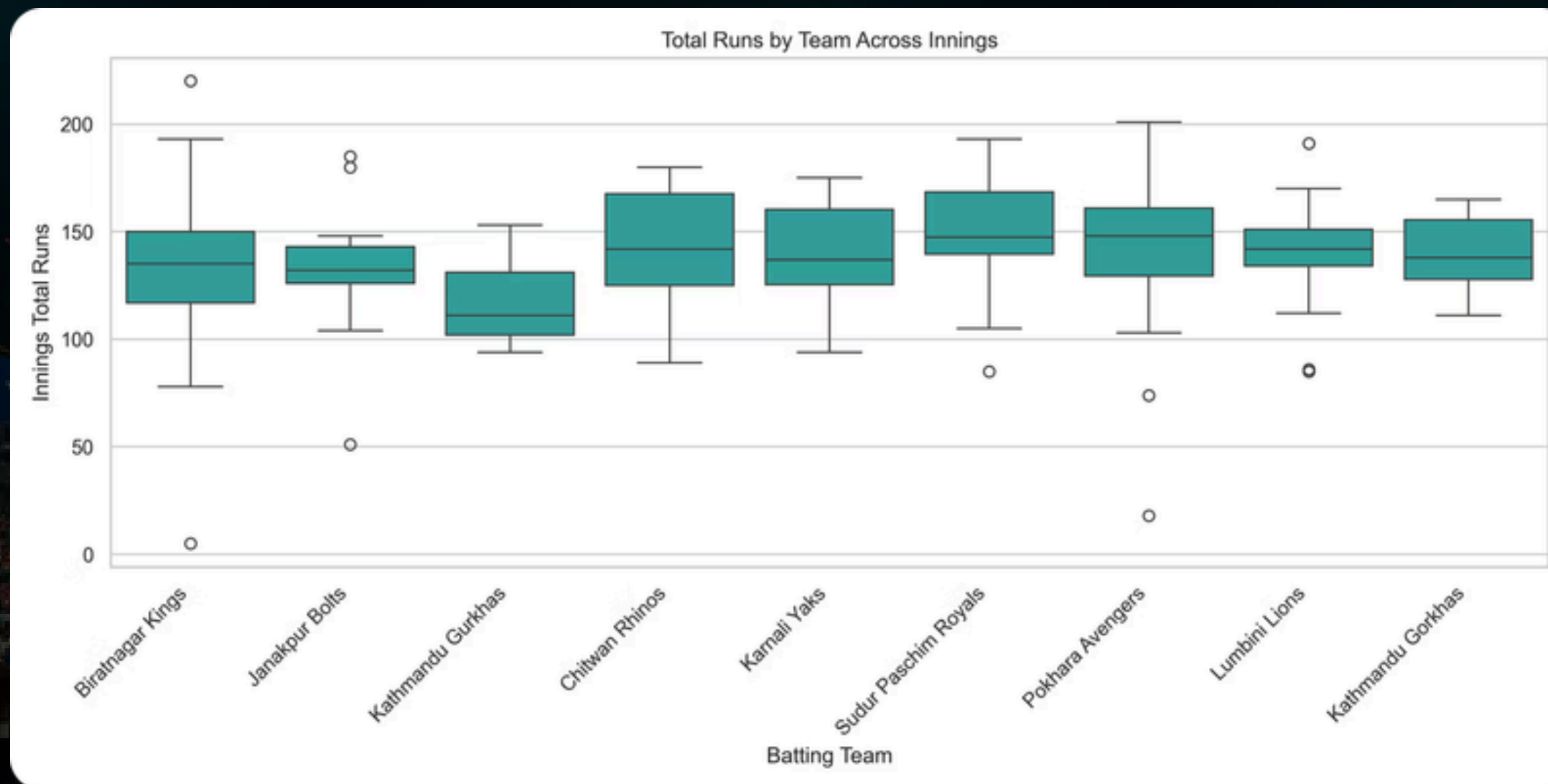


What the Histogram Reveals

First innings run totals follow a roughly bell-shaped distribution, with most matches clustering in the 140–180 range. This informs feature scaling decisions and helps identify outlier matches for further investigation.

i Understanding the baseline distribution of innings scores is critical before engineering derived features like powerplay ratios or death-over acceleration metrics.

Team Total Runs — Boxplot Analysis



Spread & Outliers

The boxplot highlights median team totals alongside interquartile ranges, making it easy to spot high-scoring outliers and low-scoring anomalies. These extremes often correlate with match-winning margins and are useful signals for the classifier.

- Median and IQR visible per team
- Outliers flagged beyond whiskers
- Useful for sanity-checking data quality

Powerplay vs. Total Runs — Scatter Plot



Early Momentum Matters

Powerplay runs show a positive correlation with total innings score — teams that start aggressively tend to post higher totals. This relationship is one of the strongest single predictors used in the logistic regression model.

Correlation does not imply causation, but powerplay performance is a reliable early indicator of match trajectory.

Feature Engineering & Modeling — model.py

Input Features (Innings 1)

Feature	Description
toss_decision_encoded	Bat=1, Field=0
powerplay_runs	Runs in overs 1–6
death_over_runs	Runs in overs 16–20
total_wickets	Wickets lost
boundary_count	4s + 6s
dot_ball_count	Scoreless deliveries

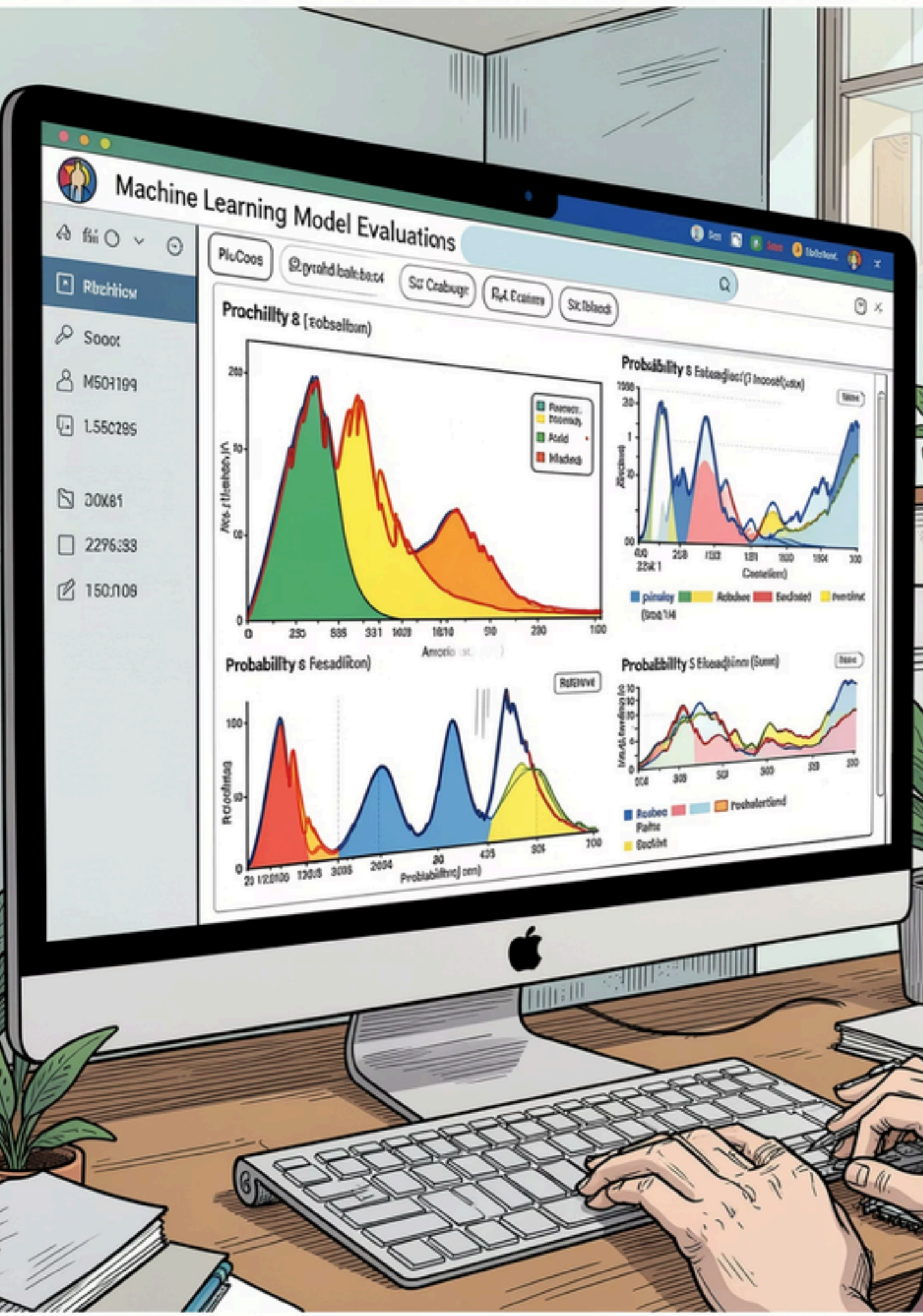
Modeling Approach

sklearn
LogisticRegression

max_iter=1000,
stratified 80/20 train-
test split.

statsmodels Logit

Provides coefficient p-
values and likelihood-
ratio test for statistical
inference.



Model Evaluation & Artifacts

All outputs are persisted to outputs/ for reproducibility and stakeholder review.



logistic_regression_model.pkl

Pickled sklearn model with metadata for downstream inference.



ROC Curve & AUC

Visualizes discriminative ability of the classifier across thresholds.



Probability Distribution

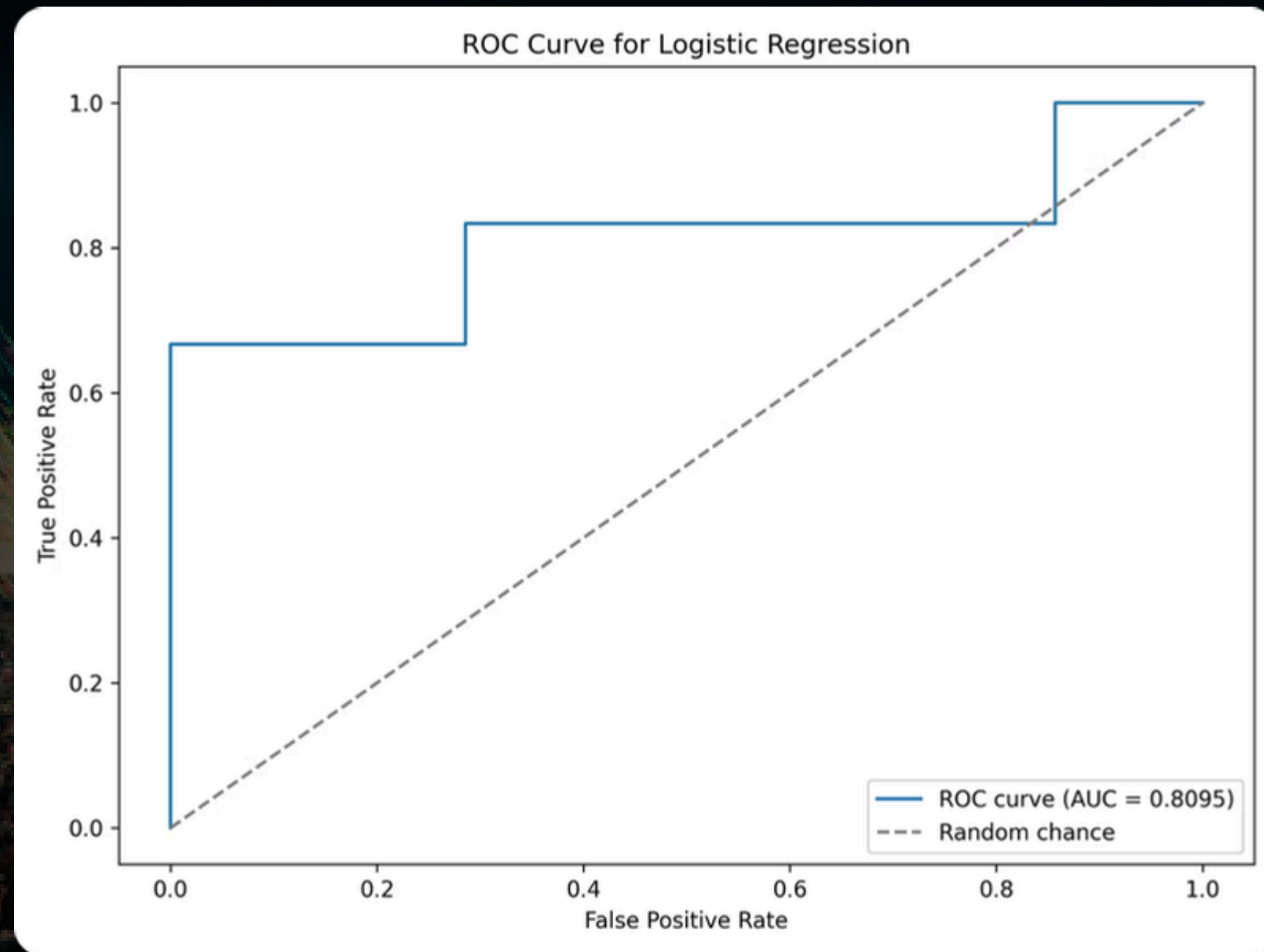
Predicted probability density by class — checks calibration and separation.



Coefficient Report

Odds ratios and statsmodels p-values printed to console for interpretability.

ROC Curve — Discriminative Performance

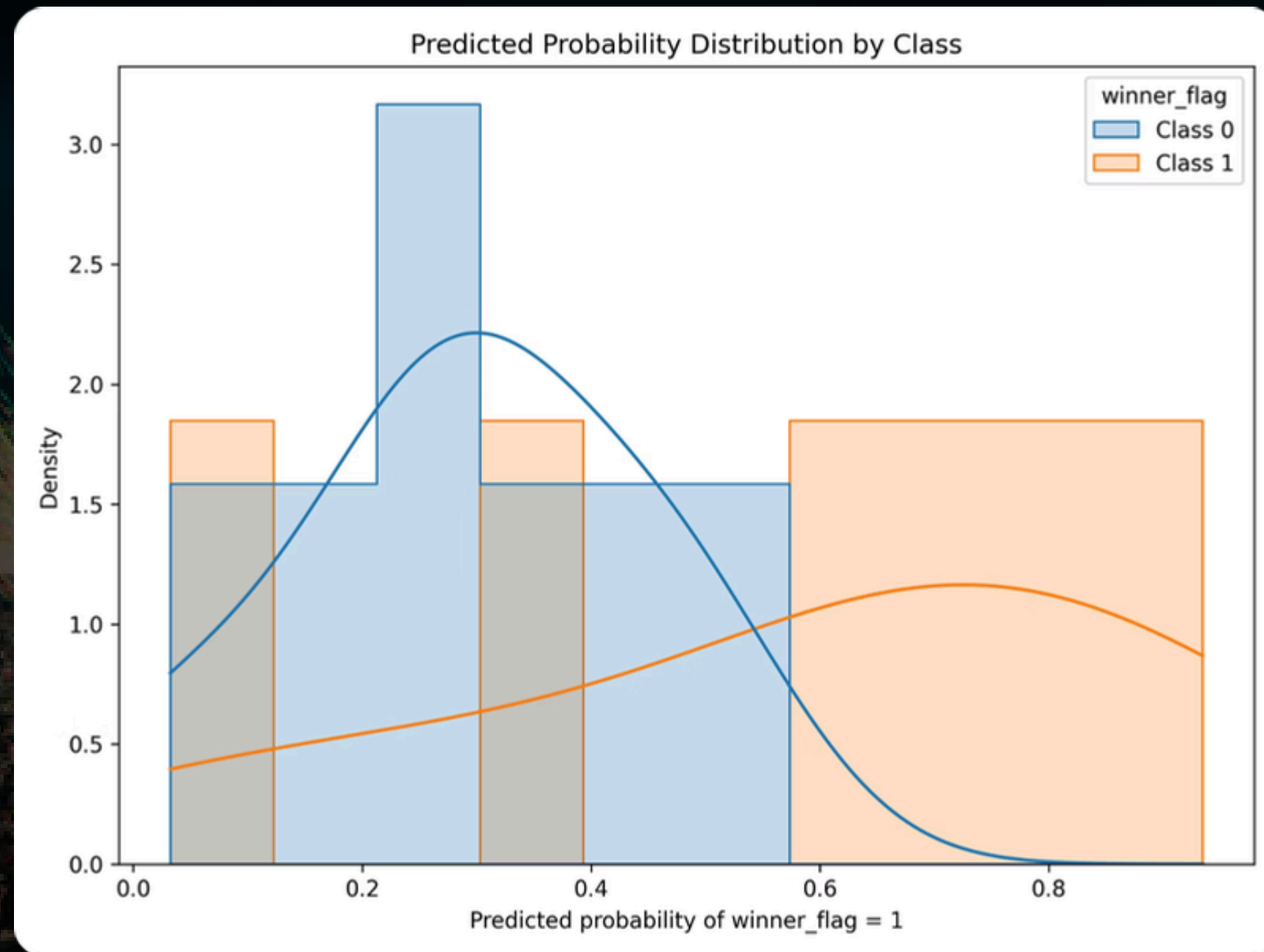


Reading the Curve

The ROC curve plots True Positive Rate vs. False Positive Rate across decision thresholds. A curve bowing toward the top-left indicates strong class separation — the model reliably distinguishes winning from losing sides.

- AUC quantifies overall discriminative power
- Threshold tuning balances precision vs. recall
- Benchmark against random classifier (diagonal)

Predicted Probability Distribution



Confidence Calibration

This histogram shows how predicted probabilities are distributed across the test set. Well-separated distributions for wins vs. losses indicate confident, accurate predictions — while overlap near 0.5 signals uncertain cases worth investigating.



Probability outputs enable risk-aware decision-making beyond binary win/loss labels.

Match inputs

Toss decision

field

Powerplay runs

22.00

Death over runs

50.00

Total wickets lost

4.00

Boundary count

22.00

Dot ball count

35.00

Predict

NPL 2025 Match Outcome Predictor

Predicts whether the batting side or bowling side is more likely to win based on match features.

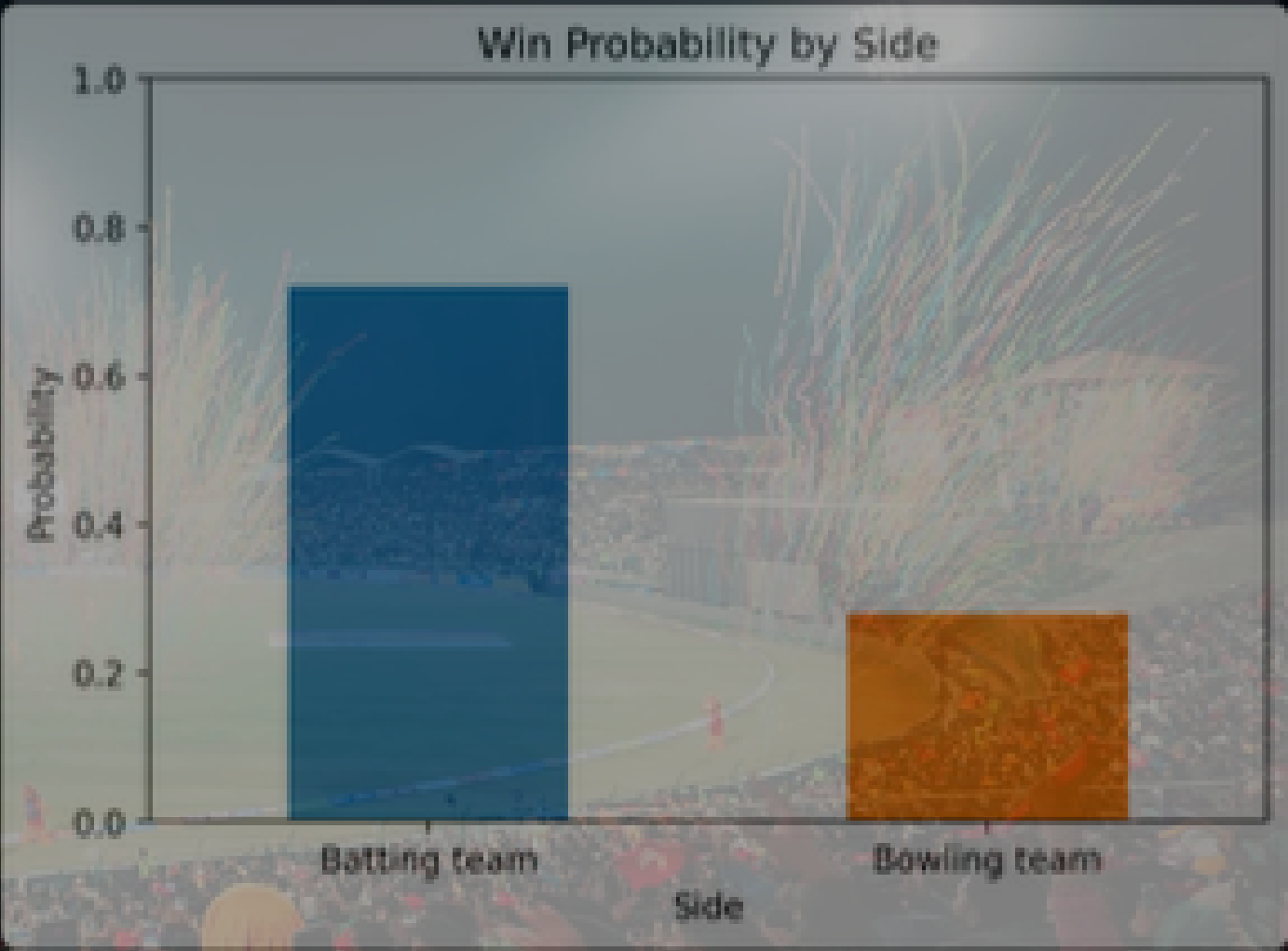
Predicted side to win

Batting team

Win probability

72.05%

DEMO



Next Steps & Improvements

Richer Features

Session-level aggregates, player statistics, and recent form indicators.

Validation Rigor

Address class imbalance; adopt time-series split for temporal realism.

1

2

3

4

Stronger Models

Random Forests, XGBoost, or neural nets with cross-validation.

Engineering Quality

Add unit tests and CI for the data pipeline.



References

Effect of Toss, Venue, and Home Advantage on T20 Match Outcomes
Reference (APA):

Winning, R., & Scanlan, A. T. (2020). Toss outcomes and home advantage in Twenty20 International cricket: A systematic examination. *International Journal of Performance Analysis in Sport*, 20(5), 749–762. <https://doi.org/10.1080/24748668.2020.1800005>

Predicting Match Outcomes in T20 Cricket Using Machine Learning

Reference (APA): Kampakis, S., & Thomas, W. (2015). Using machine learning to predict the outcome of English county twenty over cricket matches. *arXiv preprint arXiv:1511.05837*. The analytics journey transforms raw data into strategic insight through a structured and repeatable process. Each stage is essential in ensuring data becomes meaningful and actionable.

de Silva, B. M., & Swartz, T. B. (1997). Winning the coin toss and the home team advantage in one-day international cricket matches. *The New Zealand Statistician*, 32(2), 16–22. (Extended findings apply to T20 in follow-up literature)

Statistical Analysis of Performance Factors in T20 Cricket
Reference (APA): Bhattacharjee, D., & Pahinkar, D. G. (2014). Analysis of performance of bowlers using combined bowling rate. *International Journal of Sports Science and Engineering*, 6(3), 176–180.